

Bluestock Mutual Fund Analytics Platform

Final Capstone Project Report



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Bluestock Data Analyst Internship
June 2026

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1 Executive Summary

The Bluestock Mutual Fund Analytics Platform is a comprehensive financial analytics solution developed as part of the Bluestock Data Analyst Internship Capstone Project. The objective of this project was to design and implement an end-to-end analytics platform capable of ingesting, cleaning, storing, analyzing, and visualizing mutual fund industry data.

The Indian mutual fund industry has experienced substantial growth in recent years, driven by increasing investor participation, rising Systematic Investment Plan (SIP) contributions, and expanding Assets Under Management (AUM). However, investment data is often fragmented across multiple sources, making it difficult for investors and analysts to evaluate fund performance, assess risk, and identify investment opportunities.

To address these challenges, this project integrates multiple datasets including fund master information, historical NAV records, investor transactions, scheme performance metrics, SIP inflows, AUM statistics, benchmark indices, and portfolio holdings. A robust ETL pipeline was developed to clean and validate the data before loading it into a structured SQLite analytical database.

Advanced financial analytics were performed using Python and SQL, including CAGR calculations, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), Conditional Value at Risk (CVaR), and fund scorecard generation. Investor behavior analysis and cohort analysis were also conducted to identify demographic and geographic investment patterns.

The project culminated in the development of a four-page interactive Power BI dashboard that provides insights into industry trends, fund performance, investor behavior, and SIP growth. The dashboard enables stakeholders to make informed investment decisions through interactive visualizations and analytical summaries.

The final platform demonstrates the practical application of data engineering, business intelligence, financial analytics, and risk management techniques in the mutual fund domain.

2 Problem Statement and Objectives

2.1 Problem Statement

The mutual fund industry generates large volumes of financial and investor-related data on a daily basis. Investors, fund managers, and analysts require a centralized platform that can transform this raw data into actionable insights. Traditional reporting methods often lack comprehensive performance evaluation, risk assessment, and investor behavior analysis.

Furthermore, the absence of integrated analytical dashboards makes it difficult to compare funds across different categories, evaluate risk-adjusted returns, and monitor industry-wide investment trends.

This project aims to solve these challenges by building a centralized Mutual Fund Analytics Platform that combines data engineering, database management, financial analytics, and business intelligence.

2.2 Project Objectives

The primary objectives of the project are:

- Develop an automated ETL pipeline for mutual fund datasets.
- Clean, validate, and standardize multiple financial datasets.
- Design and implement a structured SQLite analytical database.
- Perform exploratory data analysis to identify industry trends.
- Calculate key financial performance and risk metrics.
- Build a fund ranking and recommendation framework.
- Analyze investor demographics and transaction behavior.
- Create an interactive Power BI dashboard for decision support.
- Generate actionable insights for investors and stakeholders.

3 Data Sources and Datasets

3.1 Overview

The project utilizes multiple datasets covering various aspects of the mutual fund ecosystem. These datasets collectively provide information regarding fund performance, investor transactions, industry trends, benchmark indices, and portfolio holdings.

3.2 Datasets Used

Dataset	Description
Fund Master	Mutual fund scheme information and metadata
NAV History	Historical Net Asset Value records
AUM by Fund House	Assets Under Management statistics
Monthly SIP Inflows	Monthly SIP contribution data
Category Inflows	Category-wise inflow and outflow statistics
Industry Folio Count	Investor participation statistics
Scheme Performance	Performance and risk metrics
Investor Transactions	Detailed transaction records
Portfolio Holdings	Scheme portfolio allocation data
Benchmark Indices	Market benchmark performance data

Table 3.1: Datasets Used in the Project

3.3 Project Scale

The platform integrates approximately:

- 10 distinct datasets
- 40 mutual fund schemes
- More than 46,000 NAV observations
- More than 32,000 investor transaction records
- Multiple years of historical financial data

These datasets provide sufficient depth for comprehensive financial and investor analytics.

4 ETL Pipeline

4.1 Overview

The ETL (Extract, Transform, Load) pipeline forms the backbone of the Mutual Fund Analytics Platform. The objective of the pipeline is to collect data from multiple sources, transform the data into a consistent and analysis-ready format, and load the processed information into a structured analytical database.

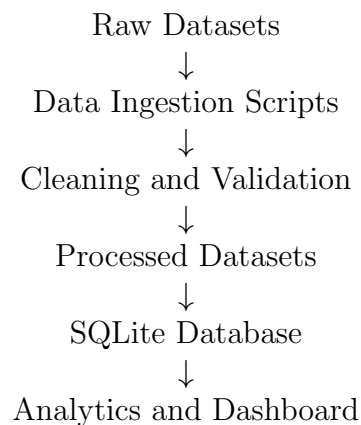
The ETL workflow was designed to minimize manual intervention while ensuring data consistency, reliability, and scalability.

4.2 ETL Workflow

The pipeline follows the following stages:

1. Data Extraction
2. Data Cleaning and Validation
3. Data Transformation
4. Database Loading
5. Analytics and Reporting

4.3 Pipeline Architecture



4.4 ETL Components

The ETL implementation consists of multiple Python scripts responsible for:

- Data ingestion
- NAV cleaning
- Transaction validation
- Performance data processing
- Database loading
- Database updates
- Analytics generation

The modular design improves maintainability and allows individual components to be updated independently.

5 Data Cleaning and Validation

5.1 Overview

Financial datasets often contain missing values, duplicate records, inconsistent formats, and invalid entries. Data quality is critical because inaccurate data can lead to incorrect analytical conclusions.

Therefore, extensive cleaning and validation procedures were implemented before loading the datasets into the analytical database.

5.2 NAV History Cleaning

The following operations were performed:

- Date standardization
- Duplicate record removal
- Missing NAV handling
- Forward-fill implementation
- NAV value validation
- Chronological sorting

5.3 Investor Transaction Cleaning

The transaction dataset was cleaned using:

- Transaction type standardization
- Date format conversion
- Amount validation
- KYC status validation
- Duplicate removal

5.4 Scheme Performance Cleaning

Performance datasets were validated through:

- Numeric value verification
- Expense ratio checks
- Return consistency validation
- Sharpe ratio anomaly detection
- Missing value treatment

5.5 Data Quality Benefits

The cleaning process significantly improved data reliability and ensured that all subsequent analytics were performed on standardized and validated datasets.

6 Database Design

6.1 Overview

A SQLite analytical database was designed to support efficient querying, aggregation, and reporting operations.

A star-schema inspired architecture was used to separate descriptive attributes from transactional and analytical records.

6.2 Database Architecture

The database consists of dimension tables and fact tables.

6.2.1 Dimension Tables

- `dim_fund`
- `dim_date`

6.2.2 Fact Tables

- `fact_nav`
- `fact_transactions`
- `fact_performance`
- `fact_aum`
- `fact_sip`

6.3 Schema Design Benefits

The selected architecture provides:

- Reduced data redundancy
- Faster aggregation queries
- Improved analytical performance
- Easier dashboard integration
- Better scalability

6.4 Analytical Queries

The database supports several business-critical queries including:

- Top funds by AUM
- Top funds by Sharpe Ratio
- Category-wise inflows
- SIP growth analysis
- Investor demographic analysis
- Transaction distribution analysis
- Benchmark comparison

The database serves as the primary source for analytical notebooks and dashboard visualizations.

7 Exploratory Data Analysis

7.1 Overview

Exploratory Data Analysis (EDA) was conducted to understand mutual fund industry trends, investor behavior, fund performance patterns, and portfolio characteristics. More than fifteen visualizations were developed to extract meaningful business insights from the processed datasets.

7.2 AUM Growth by Fund House

Assets Under Management (AUM) is a key indicator of investor confidence and fund house growth. The analysis highlights the relative market position of major fund houses.

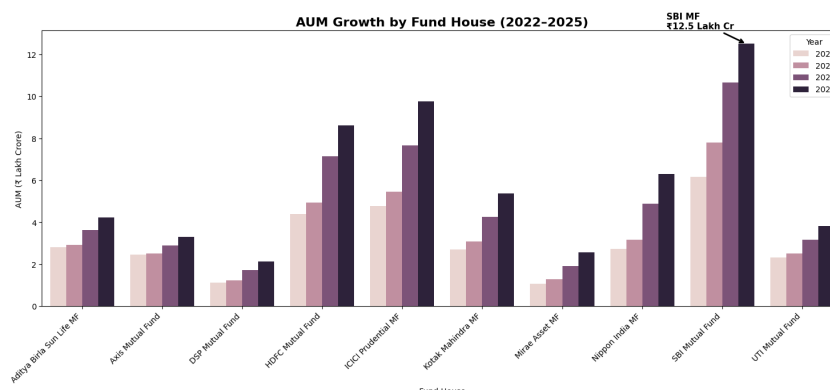


Figure 7.1: AUM Growth by Fund House

7.3 Monthly SIP Inflow Trend

Systematic Investment Plan (SIP) contributions provide insight into retail investor participation. A steady increase in SIP inflows indicates growing adoption of disciplined investing.

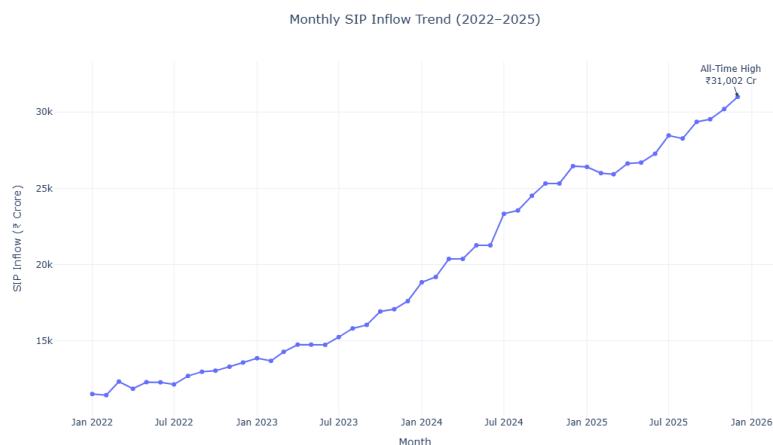


Figure 7.2: Monthly SIP Inflow Trend

7.4 Investor Age Distribution

Investor demographic analysis revealed that younger and middle-aged investors contribute a significant portion of mutual fund investments.

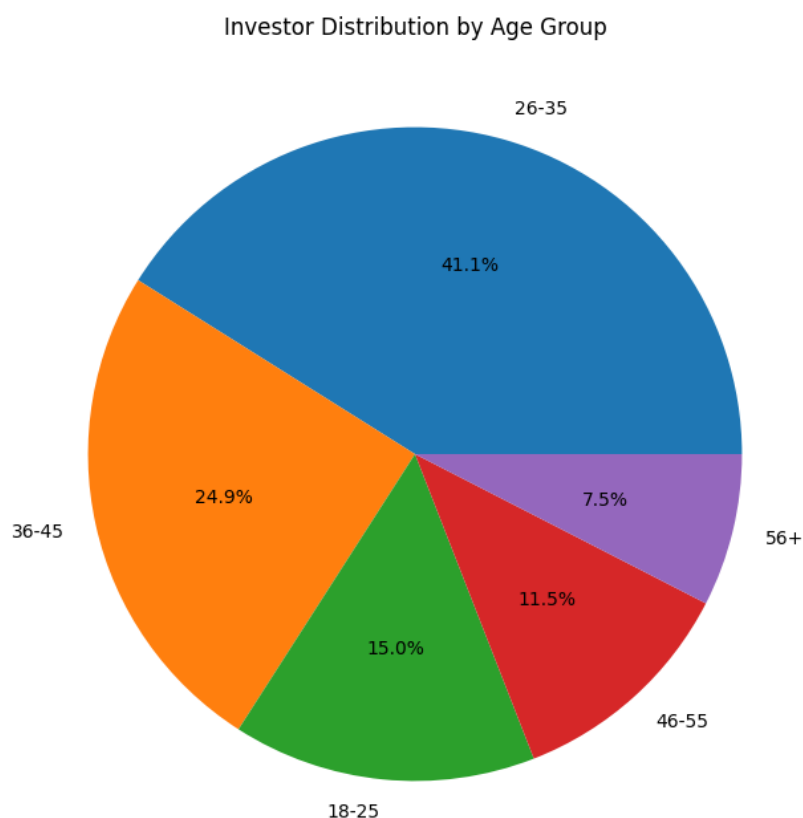


Figure 7.3: Investor Age Distribution

7.5 State-wise SIP Investment

Regional investment analysis was conducted to identify states contributing the highest investment volume.

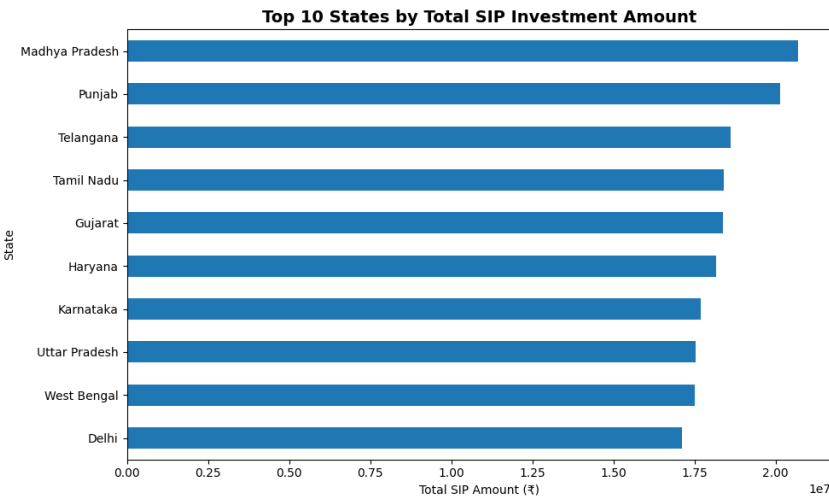


Figure 7.4: State-wise SIP Investment

7.6 Category Inflow Heatmap

Category-wise inflows were analyzed using a heatmap to identify sectors attracting higher investment activity.

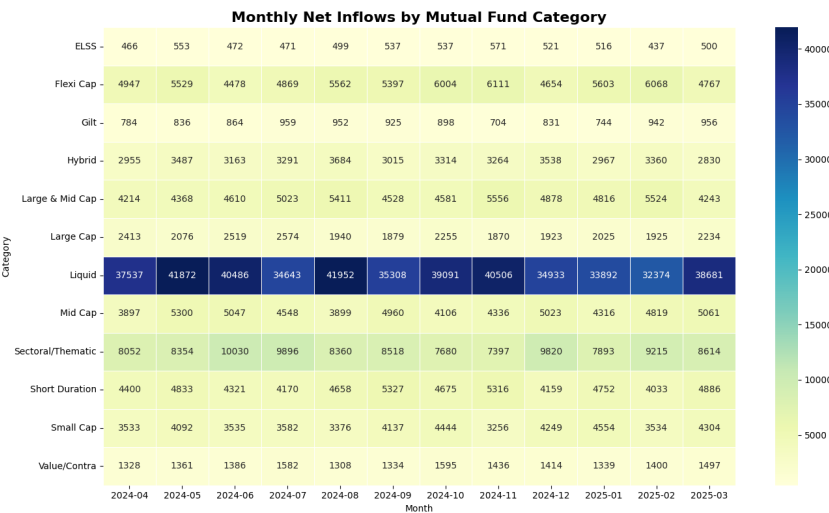


Figure 7.5: Category Inflow Heatmap

7.7 NAV Return Correlation Matrix

Correlation analysis helps identify diversification opportunities and relationships between mutual fund schemes.

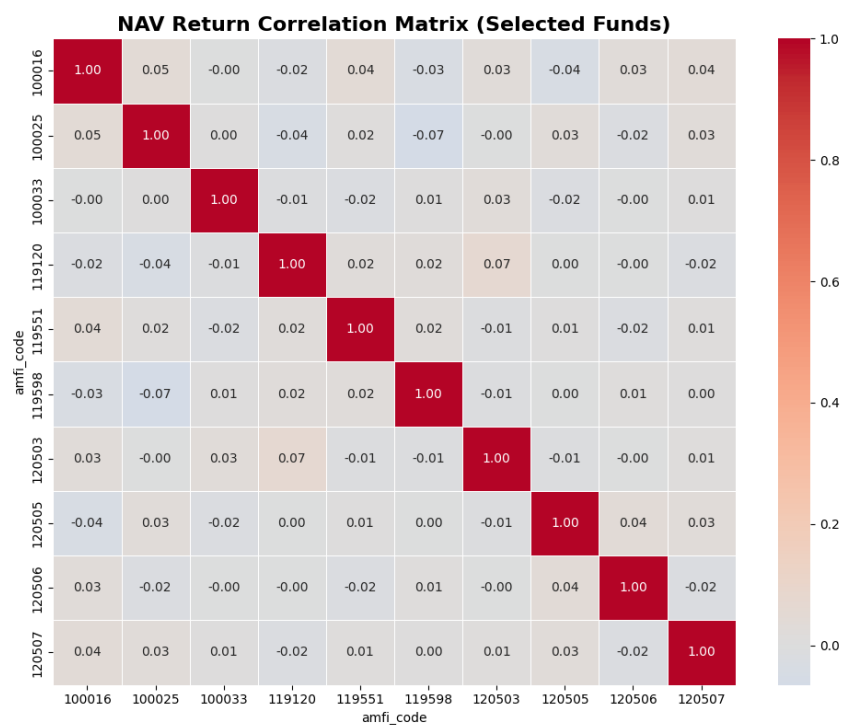


Figure 7.6: NAV Return Correlation Matrix

8 Performance and Risk Analytics

8.1 Overview

Performance analytics were conducted to evaluate the return-generating ability of mutual fund schemes while accounting for risk exposure. Multiple financial metrics were computed using historical NAV data.

8.2 Performance Metrics

The following metrics were calculated:

- Daily Returns
- 1-Year CAGR
- 3-Year CAGR
- Benchmark Returns
- Fund Scorecard Ranking

8.3 Risk Metrics

The following risk indicators were used:

- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown

8.4 Benchmark Comparison

Benchmark analysis was performed to compare fund performance against market indices and evaluate excess return generation.

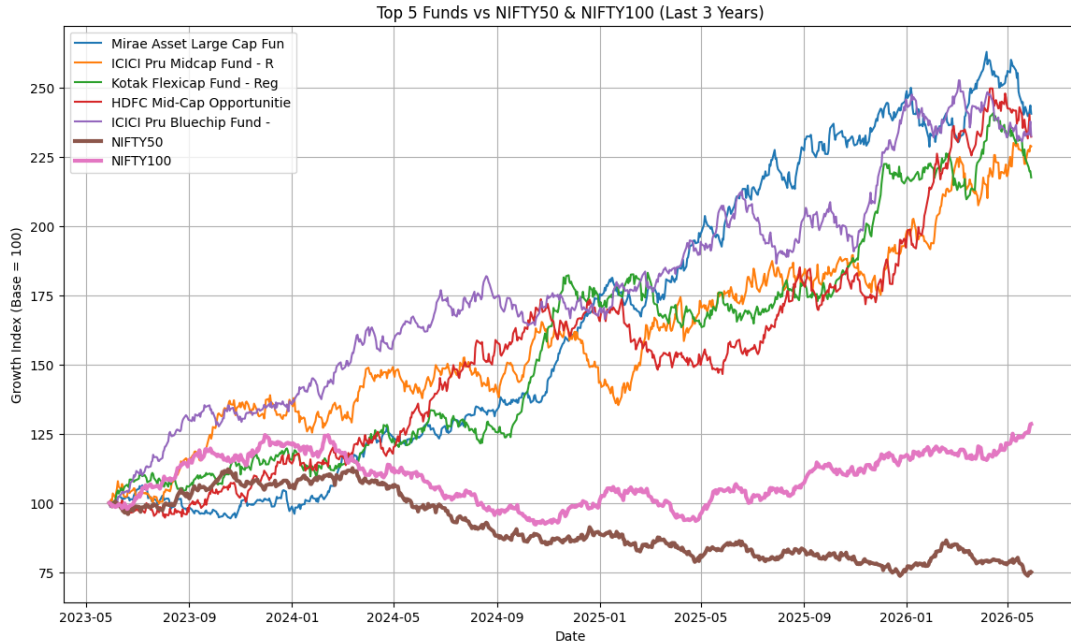


Figure 8.1: Benchmark Comparison

8.5 Fund Scorecard Methodology

A composite ranking model was developed using weighted performance and risk indicators:

- Return Ranking – 30
- Sharpe Ratio Ranking – 25
- Alpha Ranking – 20
- Expense Ratio Ranking – 15
- Drawdown Ranking – 10

8.6 Performance Insights

The analysis revealed that several funds generated strong risk-adjusted returns despite not having the highest absolute returns. This demonstrates the importance of evaluating investments using both return and risk measures.

9 Advanced Analytics

9.1 Overview

Advanced analytical techniques were implemented to provide deeper insights into portfolio risk, investor behavior, and fund selection.

9.2 Value at Risk (VaR)

Historical Value at Risk (VaR) was calculated at the 95

9.3 Conditional Value at Risk (CVaR)

Conditional Value at Risk was computed to estimate expected losses beyond the VaR threshold and provide a more comprehensive view of tail risk.

9.4 Cohort Analysis

Investor cohorts were analyzed based on transaction timing and investment activity to identify participation and retention trends.

9.5 Fund Recommendation Engine

A recommendation framework was developed to classify and suggest funds based on performance characteristics and investor risk preferences.

9.6 Sector Concentration Analysis

The Herfindahl-Hirschman Index (HHI) was used to measure portfolio concentration and diversification levels across fund holdings.

9.7 Advanced Analytics Outcome

The advanced analytics module enhanced investment decision-making by combining risk estimation, investor behavior analysis, diversification assessment, and fund recommendation capabilities.

10 Interactive Dashboard Overview

10.1 Overview

A four-page interactive Power BI dashboard was developed to transform analytical outputs into business-friendly visualizations.

The dashboard enables users to explore industry trends, evaluate fund performance, analyze investor behavior, and monitor SIP growth using interactive filters and visual analytics.

10.2 Dashboard Pages

10.2.1 Page 1 – Industry Overview

Provides industry-level KPIs, AUM trends, SIP growth trends, and fund house comparisons.

10.2.2 Page 2 – Fund Performance

Provides fund scorecards, risk-return analysis, CAGR metrics, and benchmark comparisons.

10.2.3 Page 3 – Investor Analytics

Analyzes investor demographics, geographic participation, transaction patterns, and investment behavior.

10.2.4 Page 4 – SIP and Market Trends

Tracks SIP growth, category inflows, and market participation trends.

Dashboard screenshots are included in the following pages.

11 Key Findings

- The mutual fund industry experienced consistent growth in Assets Under Management (AUM).
- SIP contributions increased steadily throughout the analysis period.
- Investors aged 26–35 represented the largest contributor group.
- Significant participation was observed from both T30 and B30 regions.
- Certain schemes generated strong risk-adjusted returns despite moderate absolute returns.
- Category-wise inflow patterns indicated varying investor preferences across market cycles.
- Correlation analysis highlighted opportunities for diversification.
- Benchmark comparison revealed differences in alpha generation across schemes.

12 Recommendations

- Investors should evaluate funds using both return and risk metrics rather than relying solely on historical returns.
- Diversification across categories can reduce portfolio concentration risk.
- SIP investing should be encouraged due to its consistent growth and long-term wealth creation benefits.
- Fund houses can target younger investor segments through digital investment initiatives.
- Geographic expansion into B30 locations can improve investor participation.
- Risk analytics such as VaR and CVaR should be integrated into investor decision frameworks.

13 Limitations

- Analysis was limited to the datasets provided during the internship project.
- Market conditions may change over time, affecting future fund performance.
- Historical performance does not guarantee future returns.
- Some investor attributes were simulated for analytical purposes.
- External macroeconomic factors were not incorporated into the analytical models.

14 Dashboard Screenshots

14.1 Dashboard Pages 1 and 2

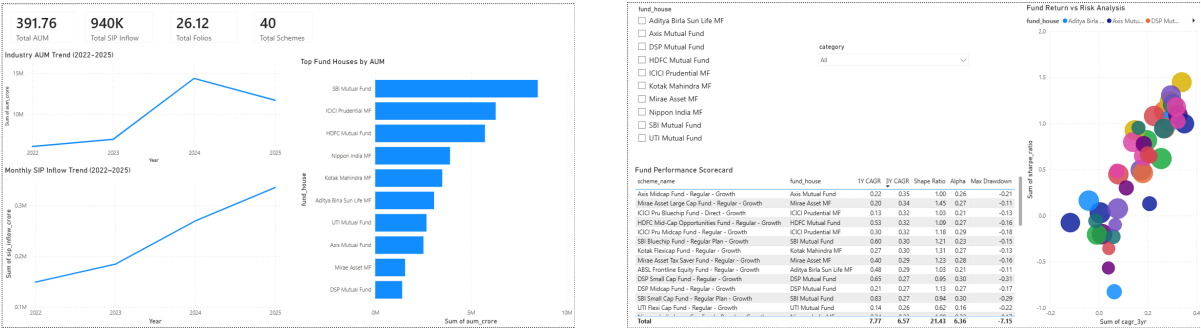


Figure 14.1: Dashboard Pages 1 and 2

14.2 Dashboard Pages 3 and 4

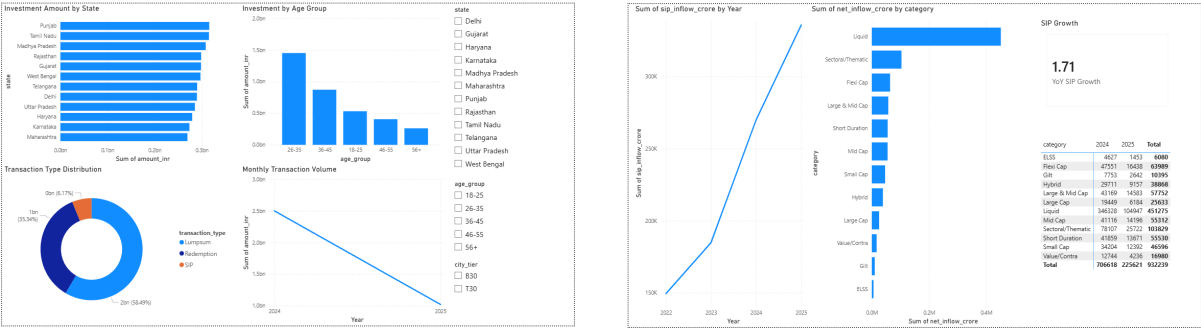


Figure 14.2: Dashboard Pages 3 and 4

15 Conclusion

The Bluestock Mutual Fund Analytics Platform successfully demonstrates the integration of data engineering, financial analytics, risk management, database design, and business intelligence within a single solution.

The project transformed multiple raw datasets into a structured analytical platform capable of generating meaningful investment insights. Through ETL pipelines, database design, exploratory analysis, performance analytics, advanced risk metrics, and interactive dashboards, the platform provides a comprehensive view of the mutual fund ecosystem.

The project highlights the practical application of data analytics in financial services and demonstrates the value of combining technical, analytical, and business intelligence skills to support data-driven investment decision-making.

— **End of Report** —